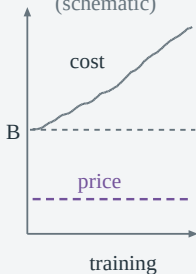


Fixed penalty

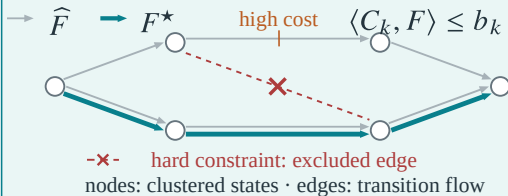
Budget leakage
(schematic)



One price rule

SafeOT

One flow, all budgets



Entropic projection

λ_k^*

Prices before violations

clip + EMA $\rightarrow \bar{\lambda}_k^*$

feedback
 $\bar{J}_k - B_k$

$$\lambda_k = \bar{\lambda}_k^* + \beta_k$$

$\tilde{A}$

PPO/TRPO

π_{t+1}

Lagrangian

Late feedback
(schematic)



Exact settings

Conditional limits

Fixed penalty

$c = 0, \beta_k$ fixed

Lagrangian

$c = 0, \beta_k$ adaptive

SafeOT

finite ε

LP

$\varepsilon \rightarrow 0$

Active-set

$\varepsilon \rightarrow \infty$
saturated clip